MYP Sciences command terms

Science skills · Grades 6, 7 and 8 · What does the question want me to do?

How to use this guide

1. In every test question, **circle the command term** (the verb: *state*, *describe*, *explain*…).
2. Find it in this guide: read **what it asks you to do** and look at the example.
3. Check the **marks**: more marks = more points to write.
4. Answer exactly what the command term asks – an *explain* answer needs a reason ("because…").

The command term ladder – one topic, five questions

	Question about an ice cube on a metal block	A good answer
State	State what happens to the ice.	It melts.
Outline	Outline what happens to the ice.	It melts quickly and turns into a puddle of water.
Describe	Describe how the ice changes.	The ice cube gets smaller and smaller; after about 4 minutes only water is left.
Explain	Explain why the ice melts quickly.	Metal is a good thermal conductor, so heat energy moves quickly from the warm block into the colder ice.
Evaluate	Evaluate using a metal tray to defrost food.	Fast because metal conducts heat (strength), but it only works if the room is warm (limitation): a good idea.

Going up the ladder, the answer needs **more detail and more thinking**.

Key

Criterion A	Knowing and understanding
Criterion B	Inquiring and designing
Criterion C	Processing and evaluating
Criterion D	Reflecting on the impacts of science
Level 1 · Remember and find	Short answers. No reasons needed.
Level 2 · Show and do	Give more detail, or do something: a graph, a plan, a calculation.
Level 3 · Think deeply	Give reasons, look at both sides, make a judgement.

Level 1 · Remember and find

Short answers. No reasons needed.

State

Criterion **A**

Give a short name, number or fact. No explanation.

Example: *State the unit of energy.*

Answer: Joule (J).

List

Criterion **A**

Give a list of short answers, one after another. No explanation.

Example: *List three renewable energy resources.*

Answer: Wind, solar, hydroelectric.

Identify

Criterion **A B**

Pick out the right answer from the information given.

Example: *Identify the independent variable in this experiment.*

Answer: The drop height of the ball.

Label

Criterion **A**

Write the names of the parts on a diagram.

Example: *Label the parts of a plant cell.*

Answer: Cell wall, cell membrane, nucleus, chloroplast, vacuole, cytoplasm.

Measure

Criterion **B C**

Find a value with an instrument (ruler, thermometer, stopwatch) and give the unit.

Example: *Measure the temperature of the water.*

Answer: 22 °C (read at eye level).

Calculate

Criterion **A C**

Work out a number answer. Show your working and give the unit.

Example: *Calculate the average of 11, 12 and 11 cm.*

Answer: (11 + 12 + 11) ÷ 3 = 34 ÷ 3 = 11.3 cm.

Draw

Criterion **A C**

Make a neat, accurate drawing or graph with a pencil and ruler: labels, units, scale.

Example: *Draw a graph of bounce height against drop height.*

Answer: Axes with labels and units, even scale, points plotted as ×, line of best fit.

Sketch

Criterion **A**

Make a quick drawing or graph that shows the main shape or features. It does not need to be exact.

Example: *Sketch how the temperature of ice changes while it melts.*

Answer: A line that goes up, stays flat while it melts, then goes up again.

Level 2 · Show and do

Give more detail, or do something: a graph, a plan, a calculation.

Annotate

Criterion **A C**

Add short notes to a diagram or graph to explain parts of it.

Example: *Annotate the graph to show where the ice melts.*

Answer: An arrow to the flat part: “ice is melting – the temperature stays at 0 °C”.

Outline

Criterion **A**

Give a short account with the main points only.

Example: *Outline how water moves in the water cycle.*

Answer: Water evaporates, condenses into clouds, falls as rain, and flows back to the sea.

Describe

Criterion A C

Give a detailed account: what it is, what happens, what you see. Not why.

Example: *Describe the trend in the graph.*

Answer: As the drop height increases, the bounce height increases steadily, from 11 cm to 53 cm.

Classify

Criterion A

Put things into groups using a rule.

Example: *Classify these materials as conductors or insulators.*

Answer: Conductors: copper, steel. Insulators: wood, plastic.

Compare

Criterion A C

Say how two or more things are alike (or different), talking about both of them.

Example: *Compare the ice melting on metal and on wood.*

Answer: On both blocks the ice melts, but on metal it melts much faster (4 min) than on wood (21 min).

Compare and contrast

Criterion A C

Give the similarities AND the differences, talking about both things.

Example: *Compare and contrast a plant cell and an animal cell.*

Answer: Both have a nucleus, membrane and cytoplasm; only the plant cell has a cell wall and chloroplasts.

Construct

Criterion C

Show information in a clear diagram, table or graph.

Example: *Construct a results table for your experiment.*

Answer: Columns: independent variable (unit), trials 1–3 (unit), average.

Present

Criterion C

Show your data so that it is easy to read (table, graph).

Example: *Present your results in a suitable way.*

Answer: A table of the trials and averages, then a bar chart because the materials are words.

Formulate

Criterion B

Write something exactly, e.g. a research question or a hypothesis.

Example: *Formulate a hypothesis for the ramp experiment.*

Answer: If I increase the ramp height, then the car goes further, because it has more gravitational potential energy.

Design

Criterion B

Plan an investigation: variables, equipment, safety and a step-by-step method.

Example: *Design an experiment to test which material insulates best.*

Answer: Same cups and water, change only the material, measure the temperature drop in 10 min, 3 trials.

Plan

Criterion B

Write the steps you will follow, in order.

Example: *Plan how you will collect your data.*

Answer: 1 set up, 2 measure the start temperature, 3 start the stopwatch, 4 read every minute, 5 repeat 3 times.

Predict

Criterion B

Say what you think will happen, before the experiment.

Example: *Predict which flask will get hottest.*

Answer: The black flask.

Estimate

Criterion A C

Find a value that is roughly right.

Example: *Estimate the bounce height for a drop from 70 cm.*

Answer: About 38 cm (read from the line of best fit).

Show

Criterion A

Give all the steps of a calculation or proof.

Example: *Show that the average of 43 and 44 cm is 43.5 cm.*

Answer: (43 + 44) ÷ 2 = 87 ÷ 2 = 43.5 cm.

Apply

Criterion A D

Use an idea you know in a new situation.

Example: *Apply what you know about insulators to keep a drink hot.*

Answer: Wrap the cup in wool: it traps air and slows heat transfer.

Suggest

Criterion B C D

Give a possible answer or idea. There can be more than one right answer.

Example: *Suggest an improvement to the method.*

Answer: Film the bounce in slow motion to read the height more accurately.

Deduce

Criterion A C

Reach a conclusion from the information given.

Example: *Deduce whether the solution is an acid or an alkali.*

Answer: Universal indicator turned red, pH 2, so it is an acid.

Solve

Criterion A

Find the answer to a problem using a method or formula.

Example: *Solve: how far does a car go at 2 m/s in 5 s?*

Answer: distance = speed × time = 2 × 5 = 10 m.

Use

Criterion A

Use your knowledge or a rule to get an answer.

Example: *Use the graph to find the bounce height at 60 cm.*

Answer: Go up from 60 cm to the line, across to the axis: about 33 cm.

Interpret

Criterion A C

Use what you know to find trends and conclusions in data.

Example: *Interpret the bar chart of the metal rods.*

Answer: Copper's pin dropped first, so copper is the best conductor; steel is the worst.

Summarize

Criterion A C

Give the main points in a short form.

Example: *Summarize your results.*

Answer: The higher the drop, the higher the bounce; the pattern is a straight line through 0.

Level 3 · Think deeply

Give reasons, look at both sides, make a judgement.

Explain

Criterion **A B C D**

Give a detailed account with reasons: say WHY or HOW, often with “because”.

Example: *Explain why the ice melts faster on metal.*

Answer: Metal is a good thermal conductor, so heat energy moves from the block into the ice quickly.

Analyse

Criterion **A C**

Break information into parts to find the important points, patterns and links.

Example: *Analyse the results table.*

Answer: Two rows follow the pattern, 61 cm is an anomaly, and the averages double when the height doubles.

Discuss

Criterion **C D**

Look at different points of view or factors, with evidence, then come to a conclusion.

Example: *Discuss the use of wind turbines in your town.*

Answer: Clean and renewable, but noisy and not always windy; overall a good addition to other energy resources.

Evaluate

Criterion **C D**

Judge something by weighing up its strengths and limitations.

Example: *Evaluate the validity of your method.*

Answer: Fair test with 3 trials (strength), but reading by eye and one anomaly (limitations): mostly valid.

Justify

Criterion **B C D**

Give good reasons or evidence for your answer.

Example: *Justify your choice of a bar chart.*

Answer: The independent variable is in words (types of metal), so a bar chart is correct, not a line graph.

Comment

Criterion **C**

Give your judgement about a result or statement.

Example: *Comment on the reliability of the results.*

Answer: The trials in each row are close together, so the results are reliable.

Investigate

Criterion **B**

Study something carefully and step by step to find out the facts.

Example: *Investigate how the drop height affects the bounce height.*

Answer: Plan, do the experiment with 3 trials per height, process the data and conclude.

A–Z: find a command term quickly

Term	Level	Criterion
Analyse	Level 3	A C
Annotate	Level 2	A C
Apply	Level 2	A D
Calculate	Level 1	A C
Classify	Level 2	A
Comment	Level 3	C
Compare	Level 2	A C
Compare and contrast	Level 2	A C
Construct	Level 2	C
Deduce	Level 2	A C
Describe	Level 2	A C
Design	Level 2	B
Discuss	Level 3	C D
Draw	Level 1	A C
Estimate	Level 2	A C
Evaluate	Level 3	C D
Explain	Level 3	A B C D
Formulate	Level 2	B
Identify	Level 1	A B
Interpret	Level 2	A C
Investigate	Level 3	B
Justify	Level 3	B C D
Label	Level 1	A
List	Level 1	A
Measure	Level 1	B C
Outline	Level 2	A
Plan	Level 2	B
Predict	Level 2	B
Present	Level 2	C
Show	Level 2	A
Sketch	Level 1	A
Solve	Level 2	A
State	Level 1	A
Suggest	Level 2	B C D
Summarize	Level 2	A C
Use	Level 2	A

Command terms are the verbs the IB uses in MYP Sciences questions. The meanings here are written in simple English for Grades 6–8; the examples come from our units (energy, water, nature, acids, forces and others).